

Test Management Assignment

Important Notes for Test Management Assignment:

This assignment is divided into two parts, focusing on the analysis of the testing process and release quality of a project's upcoming release. Read the background information carefully, and answer the questions.

Submission: As a PDF file.

Grading:

Question 1:

1.5p: The answer thoroughly analyses the information in the figure and gives a sensible interpretation. The answer contains a well motivated recommendation regarding further testing.

1-0.5p: The answer is incomplete with respect to the information analysed or the recommendation regarding further testing, it does not provide adequate motivation, or it is not entirely accurate compared to the information given.

0p: The answer is very incomplete, lacks proper motivation, or has severe inaccuracies compared to the information given.

Question 2:

1p: The similarities and differences mentioned are correct and well explained using the information given.

0.75-0.25p: Some similarities and differences are accurate, but the explanation lacks completeness or may not entirely derive from the information given.

0p: The similarities and differences are incorrect or not explained through the information given.

Question 3:

1.5p: The answer gives a clear yes or no recommendation, it has a comprehensive and well-motivated explanation, thorough discussion of risks, it contains a solid mitigation plan, and effective linkage to both previous and current iteration data.

1-0.5p: The answer gives a clear recommendation, generally well-motivated explanation, but may lack depth in certain areas. Risks and mitigation are discussed, but some aspects could be more detailed. There is adequate linkage to available data.

0p: The response lacks a clear yes or no recommendation, has an insufficient explanation, or fails to address risks, mitigation, and data linkage.

Use of AI and Collaboration with Other Students: Using AI tools to produce an answer for this assignment is not permitted. All parts of the submission must be produced by yourself (apart from files provided in the assignment). Collaboration with other students is not permitted. This is an individual assignment that you must author independently.

Part 1: Analysing the testing process

Imagine you are the Test Manager at Stylebox company. Now, the company prepares to release a new version of its online web shop system. The new release touches every aspect of the system, including the customer-facing store frontend, all backend systems, and tools for all internal stakeholders. The new release will also be deployed in new markets and a large number of new customers and service personnel will start using the system. There has been a lot of preparation for the system and the marketing campaign is assuming that the release will go out on schedule. *At this time, it is one week from code freeze, and the freeze period is one week. In other words, the new release will go live in two weeks.*

The company's software development uses an agile software development process where they can do testing simultaneously with development and they reserve one week of time at the end of each sprint where they stop adding new features and concentrate on fixing defects only – this is called the code freeze period.

Your task in the analysis is to identify problems in the testing process and recommend ways in which it could be improved.

Start by familiarising yourself with the topic in Chapter 4.2 of Kan (2003) – see [Aalto Primo](#) for a link to the online book. The key concept of Defect arrival during testing refers to defects that have been found through testing. In other words, when a defect is found by testing, we say that the defect "arrives" – and we record these arrivals to count the Defect arrival rate.

Given the defect arrival pattern shown in the figures below, answer the following question. (The figures show no numbers. This is on purpose, to focus the analysis on the shape of the curve.)

Question1: What does the pattern in Figure 1 tell about how the testing effort was timed? Should more testing be conducted? Motivate your answer and describe how it relates to what is shown in the figure.

For this question, there is no specific requirement regarding the length of your response. However, it is crucial to ensure that your answer comprehensively covers each part of the question and that you motivate your answer by referring to information in Figure 1. There can be multiple different interpretations of the data. The important thing is that you explain your reasoning and the assumptions behind your answer. You can also discuss different possibilities and the reasoning behind them. You can write your answer in essay format or utilise the provided format for clarity and structure:

1. Week Interval (A-B): <Your observations and insights go here>
2. Week Interval (C-D): <Your observations and insights go here>
3. ...

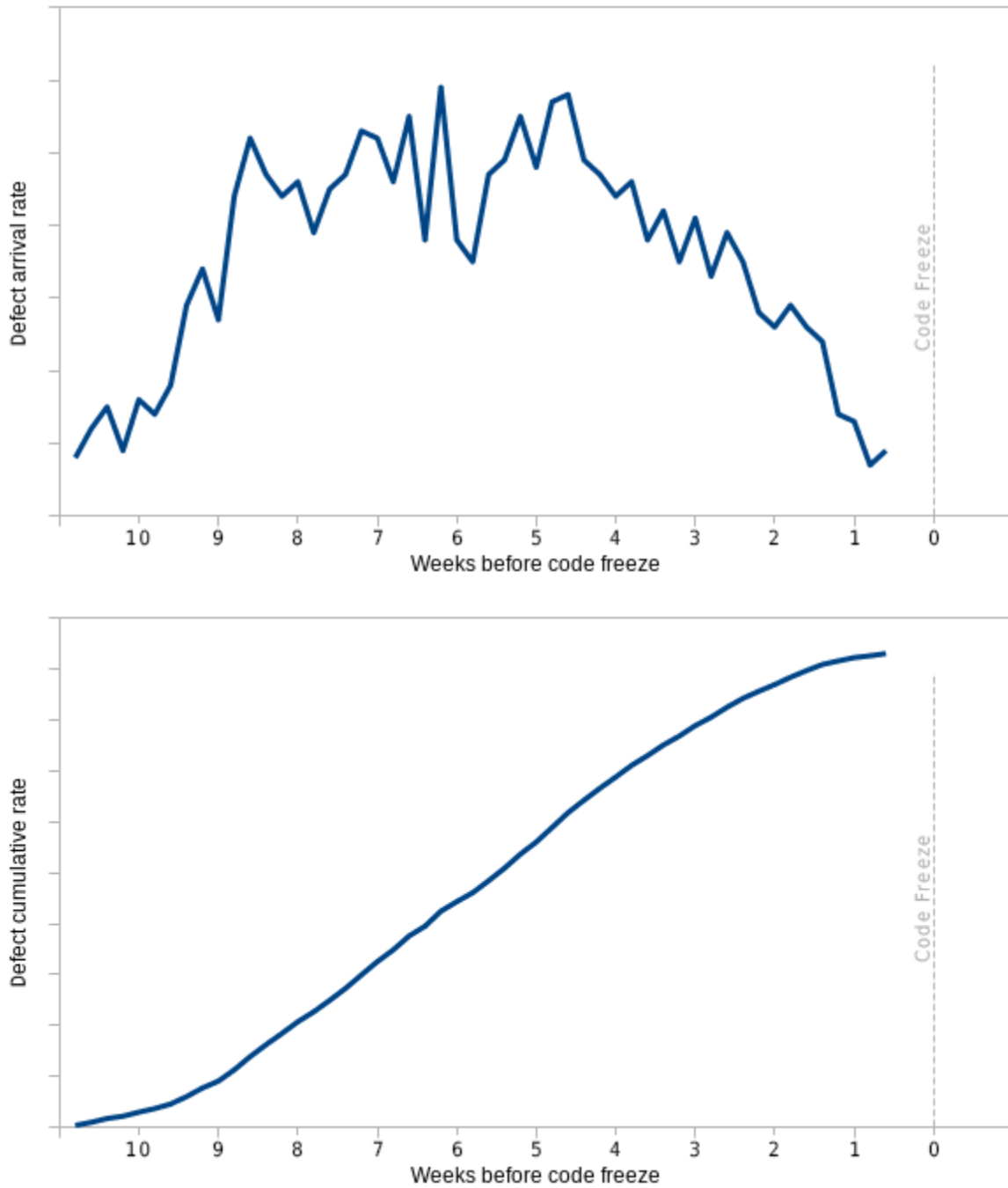


Figure 1. Defect arrival rate (top) and cumulative arrival rate (bottom) for the previous release.

Part 2: Can we release?

In this part, as the test manager of the Stylebox company, you will further analyse the testing process in preparation for a new version of its online web shop system. As a reminder, the company is preparing a release that touches every aspect of the system, including the customer-facing store frontend, all backend systems, and tools for all internal stakeholders. The new release will also be deployed in new markets and a large number of new customers and service personnel will start using the system. There has been a lot of preparation for the system and the marketing campaign is assuming that the release will go out on schedule. It is one week from code freeze, and the freeze period is one week. In other words, the new release will go live in two weeks.

You have just presented your findings from the previous assignment to the management team. The CTO tasks you with a new assignment: to give a release recommendation.

You can recap the topic in Chapter 4.2 of Kan (2003). Also, familiarise yourself with Chapter 10.4 in Kan (2003). You can find a link to the online book in [Aalto Primo](#).

You obtain the defect arrival data for the previous release (Figure 1). You learn from the head of testing that new unit tests were added since then that kept the testing coverage at the same level. The head of testing says: "We continued with the current system and acceptance tests because the user tasks stay the same in the next release. The acceptance specs didn't change, so we didn't change anything in the integration, system, or acceptance tests."

Question2: When comparing the patterns in figures 2 and 3, what does this tell about the release quality of the new, upcoming release? Describe your observations in terms of the figures.

For the question, there is no specific requirement regarding the length of your response. However, it is crucial to ensure that your answer comprehensively covers each part of figure 3, which contains new information and that you motivate your answer by referring to and comparing information in figures 2 and 3. You must not only provide an overall analysis of where the curve ends up at the end, but also discuss different time intervals within the figures and explain what could have caused the effects you see. There can be multiple different interpretations of the data. The important thing is that you explain your reasoning and the assumptions behind your answer. You can also discuss different possibilities and the reasoning behind them. You can write your answer in essay format or utilise the provided format for clarity and structure:

Week Interval (A-B): <Your observations and insights go here>

Week Interval (C-D): <Your observations and insights go here>

...

Question3: You present your findings in a follow-up meeting with the management team. They ask: Our big launch is approaching. Should we release the new software version in two weeks? What is your answer? Remember that the management team needs a yes or no answer, but they also want to know why, what the risks involved in both options are, and what mitigation approach you would propose for those risks for your recommended option. Motivate your answer and link the motivation to the data you have available. Remember: you have data from the previous iteration and the current one – use both!

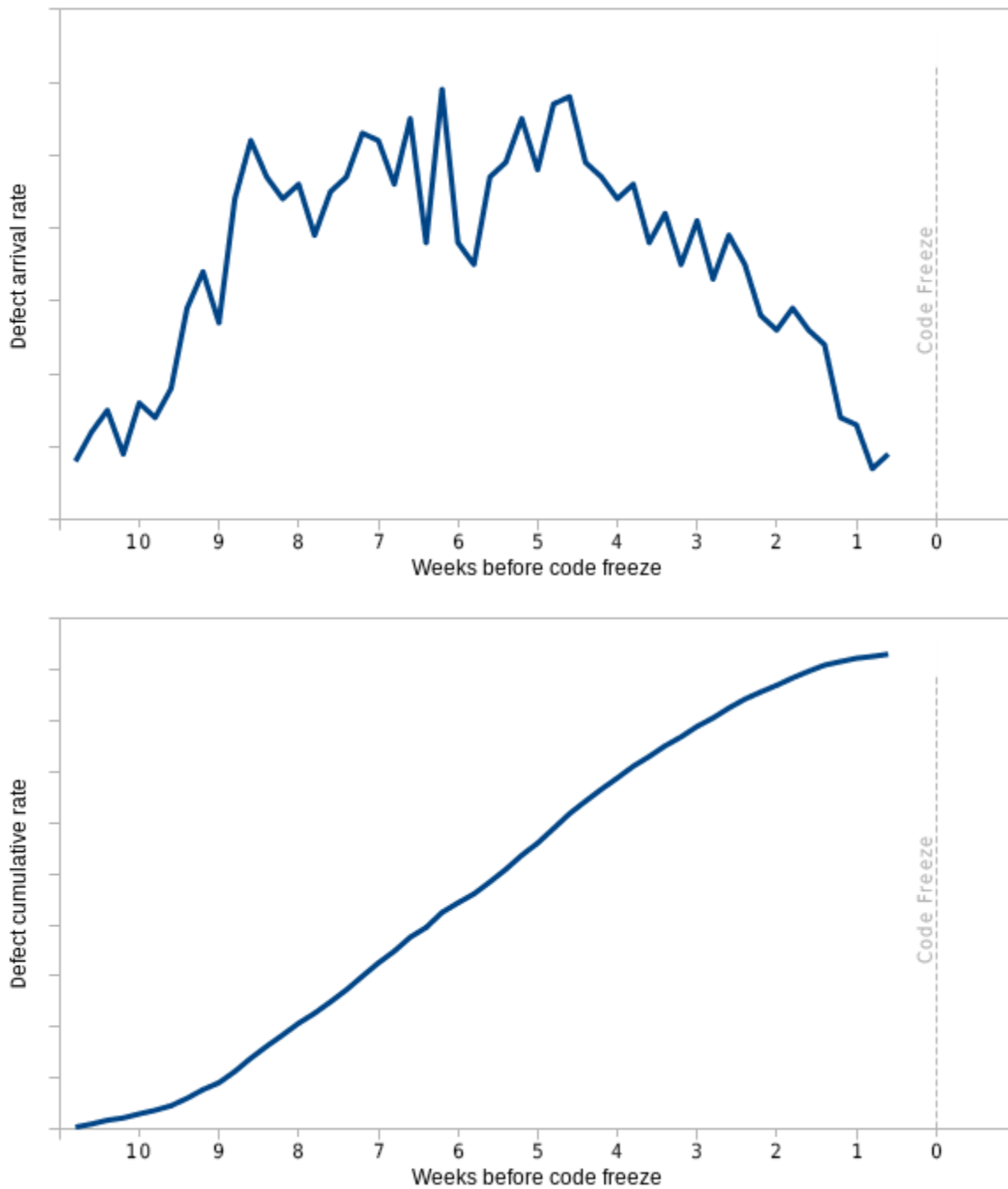


Figure 2. Defect arrival rate (top) and cumulative arrival rate (bottom) for the previous release.

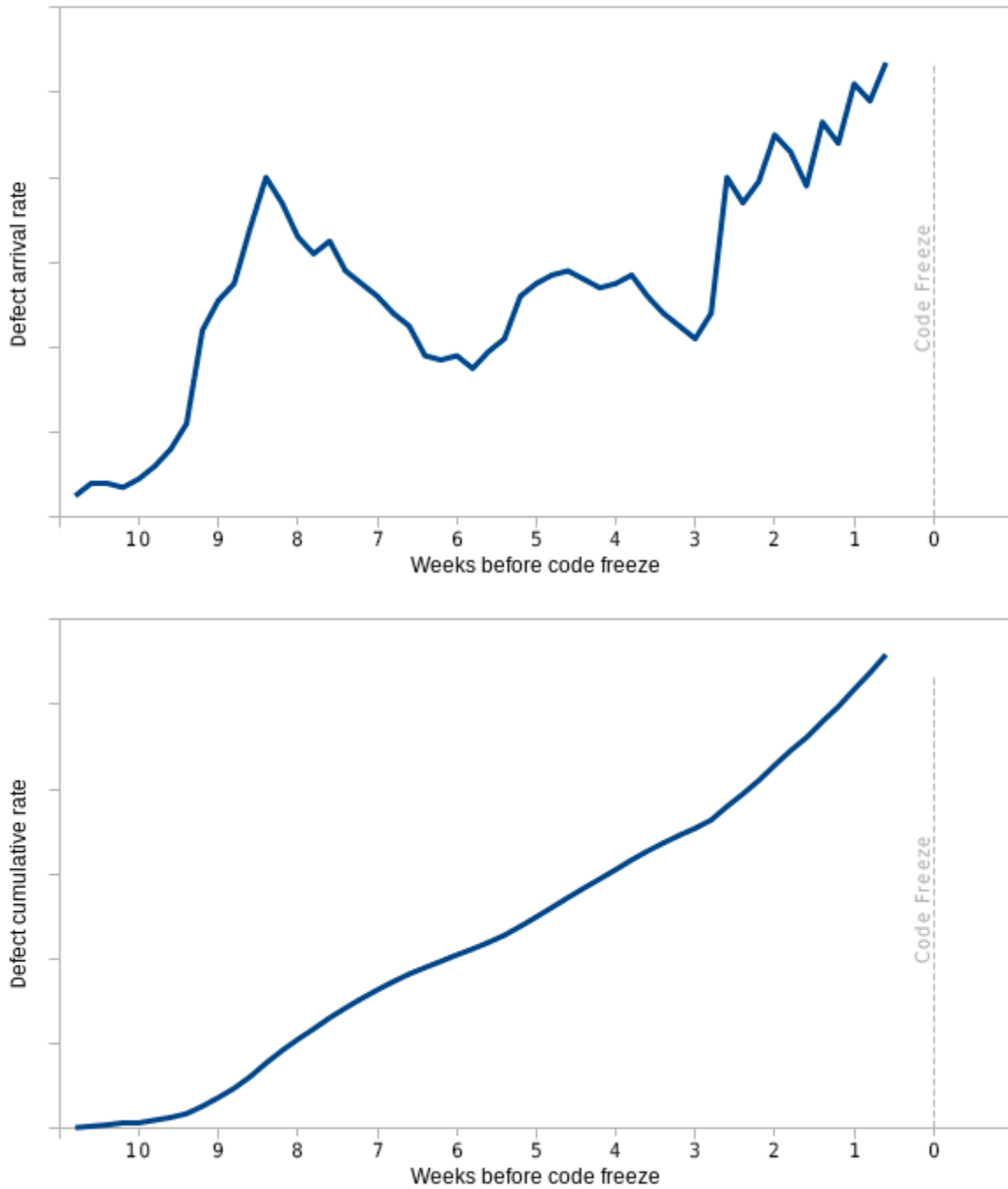


Figure 3. Defect arrival rate (top) and cumulative arrival rate (bottom) for the next release.